Identifying Causes of Protectionism Using Double Machine Learning

Emily Harms · POL380 Paper Proposal

1 Proposed Topic

Why do some democracies protect domestic producers more than others? I aim to study whether factors such as exports, unemployment rates, GDP, and inflation rates causally shape how protectionist a country's trade policy is. Other analyses (Ehrlich, 2007; Gawande and Bandyopadhyay, 2000; Henisz and Mansfield, 2006) often test for the impact of interactions between firms and policymakers and domestic institutions on protectionism, while holding economic conditions constant, meaning that additional testing must be done in order to determine whether the economic conditions themselves also impact the degree of protectionism. Using a panel of many countries spanning several decades, I aim to estimate the effect of each economic factor using double machine learning (DML), which lets flexible learners such as random forests and neural networks estimate causal parameters in the presence of complex confounding variables (Chernozhukov et al., 2016).

This project aims to answer the question **Which domestic economic factors cause countries to adopt more protectionist trade policies, and how robust are these effects?** by testing a variety of hypotheses regarding protectionism.

H1 (Exports): Exports being a higher percentage of GDP decrease protectionism. Countries that export a significant amount of products abroad prefer other countries to keep their markets open, and worry that raising their own tariffs may lead to retaliation.

H2 (Unemployment): Increased unemployment rates cause increased protectionism. Free trade allows products to be imported from abroad, meaning that local industries may suffer, leading to high unemployment rates. Therefore, those suffering from unemployment and the firms forced to lay off workers will advocate for protectionism in order to maintain local jobs.

H3 (GDP per Capita): Increases in GDP per capita causes increased protectionism. need help with this one

H4 (Total GDP in US$): A higher total GDP causes increased protectionism. Small economies are unable to produce all goods, therefore imports are essential. Larger economies are able to produce more goods domestically and can afford for trade to be closed Is this right? Check if Ehrlich cited anything useful and look at that paper.

H5 (Inflation): Increased inflation rates cause decreased protectionism. Inflation is defined by an increase in domestic prices. Decreased protectionism allows for the importing of cheaper goods, helping to reduce prices.
For the sake of this analysis, the level of protectionism will be quantified using tariff rates. Tariffs are directly set by policymakers, meaning that changes in tariff rates directly reflect policy choices.

2 Prior Work

Prior analyses argue that politics drives protectionism, rather than economics. In the Grossman-Helpman model (Grossman and Helpman, 1994), governments weigh campaign contributions from organized industries against social welfare. Gawande and Bandyopadhyay (2000) tested this on US non-tariff barriers across manufacturing industries and found broad support for the model. A second set of studies ask how political institutions affect these pressures. Ehrlich (2007) finds that a larger number number of electoral districts, a larger number of political parties, presidentialism, and left-wing governments all cause increased protectionism. Henisz and Mansfield (2006) focuses on how institutions change the effect of economic conditions. All of these studies focus on the impacts of interactions between lobbyists and policymakers, and/or the institutions involved. This suggests that more research needs to be done into the impact of the economic conditions themselves. Do the economic conditions that drive these interactions also have a role, or are the interactions the root cause? This project seeks to study the impact of various economic conditions on the level of protectionism.

3 Proposed Methods and Current Progress

Double Machine Learning (DML) is based on the Frisch-Waugh-Lovell (FWL) Theorem (Waugh and Frisch, 1933; Lovell, 1963), which says that interactions between variables can be combatted using multiple linear regression analyses. First, regress the treatment parameter D on the independent variable you'd like to test (X_T) to find the treatment residual (r_T). Then, regress the outcome (Y) on the nuisance variables (X_N) to obtain the nuisance residual (r_N), giving you the variation of the outcome independent of X_T. Finally, regress r_N on r_T to obtain the multivariate regression coefficient. DML takes the FWL Theorem a step further, expanding it beyond linear relationships. DML replaces linear regression with any machine learning algorithm, enabling the FWL process to be used on models in which the variables are not related linearly.

Add a paragraph on how prior analyses use linear regression. Mention the assumption about linear relationships and use as justification for DML. Also I don't think Ehrlich even attempted FWL, so that's not good either.

Data on tariff rates (World Bank Group, 2026*e*), exports (World Bank Group, 2026*a*), unemployment rates (World Bank Group, 2026*f*), GDP (World Bank Group, 2026*b*,*c*), and inflation (World Bank Group, 2026*d*) and the number of political parties (Varieties of Democracy, 2026) have been obtained, and the required data reduction and cleaning is complete. A simple Python program (Appendix A) was written to pull the data needed from each dataset, and add the data to one large .csv file. A subset of the data is shown in table 1

Country	Year	% Exports	Unemp.	GDP/Capita	GDP (US$)	Inflation	No. Parties	Tariff
ALB	1997	10.04	16.874	728.55	4.71e9	33.18	2.87	14.41
ALB	2000	19.53	19.023	1160.42	6.15e9	0.05	2.87	11.65
...	...	...	...	...	...	...	...	...
ZMB	2020	46.72	6.032	951.64	2.34e10	15.73	2.78	4.82

Table 1: Dataset that will be used for double machine learning. % Exports, unemployment rate, GDP/capita, GDP in US$, inflation rate, and number of political parties are independent variables, tariff rate is the dependent variable.

Parameter	coef	std err	t	$P > \|t\|$	2.5 %	97.5 %
Exports	-0.0227	0.00962	-2.363	0.0181	-0.0416	-0.00388
Unemployment	0.0279	0.0511	0.546	0.585	-0.0722	0.128
GDP/Capita	-3.898	8.559e-6	-4.554	5.254e-6	-5.483e-5	-2.221e-5
GDP (US$)	-2.473e-13	0.00153	-1.610	0.107	-5.483e-13	5.375e-14
Inflation	-5.581e-4	0.00153	-0.366	0.715	-0.00355	0.00243
Number of Parties	0.0679	0.0725	0.936	0.350	-0.0743	0.210

Table 2: Preliminary results. A P-value less than 0.05 shows a statistically significant causal relationship.

4 Impacts of Protectionism

References

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World Bank Group. 2026*c*. "GDP per capita (current US$) [NY.GDP.PCAP.CD]." World Development Indicators.

World Bank Group. 2026*d*. "Inflation, consumer prices (annual %) [FP.CPI.TOTL.ZG]." World Development Indicators.

World Bank Group. 2026*e*. "Tariff rate, applied, weighted mean, all products (%) [TM.TAX.MRCH.WM.AR.ZS]." World Development Indicators.

World Bank Group. 2026*f*. "Unemployment (% of total labor force) [SL.UEM.TOTL.ZS]." World Development Indicators.

A Data Cleaning Code

```
# Emily Harms
# POL380 Term Project: Causes of Protectionism
# Clean data from various sources to get consistent/usable formatting

# import required modules

import pandas as pd
import numpy as np

# Start with tariff data

TariffData = pd.read_csv('TariffData.csv', skiprows = 4)

TariffData = TariffData.drop(columns = ['Country Name', 'Indicator Name',
    'Indicator Code']) # Get rid of columns we don't need

# Make a new DataFrame, where I will put the data I want to keep

ProtectionismData = pd.DataFrame()

# Make lists where I will store tariff data, for now

countries = []
years = []
tariffs = []

# Fill lists with tariff data

for row in range(len(TariffData)):
    for col in range(1, TariffData.shape[1]):
        tariff = TariffData.iloc[row, col]
        if pd.notna(tariff):
            countries.append(TariffData.iloc[row, 0])
            years.append(TariffData.columns[col])
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            tariffs.append(tariff)

# Save data to the big dataset

ProtectionismData['Country'] = countries
ProtectionismData['Year'] = years
ProtectionismData['Tariff'] = tariffs

# Clean and add export data

ExportData = pd.read_csv('ExportData.csv', skiprows = 4)

ExportPercent = []

for i in range(len(ProtectionismData)):
    code = ProtectionismData.loc[i, 'Country']
    year = str(ProtectionismData.loc[i, 'Year'])
    row = ExportData[ExportData['Country Code'] == code]
    if len(row)>0 and year in ExportData.columns:
        ExportPercent.append(row[year].iloc[0])
    else:
        ExportPercent.append(np.nan)

ProtectionismData['Exports'] = ExportPercent

ProtectionismData = ProtectionismData.dropna(subset =
    ['Exports']).reset_index(drop=True)

# Clean and add unemployment data

UnemploymentData = pd.read_csv('UnemploymentData.csv', skiprows = 4)

UnemploymentPercent = []

for i in range(len(ProtectionismData)):
    code = ProtectionismData.loc[i, 'Country']
    year = str(ProtectionismData.loc[i, 'Year'])
    row = UnemploymentData[UnemploymentData['Country Code'] == code]
    if len(row)>0 and year in UnemploymentData.columns:
        UnemploymentPercent.append(row[year].iloc[0])
    else:
        UnemploymentPercent.append(np.nan)

ProtectionismData['Unemployment'] = UnemploymentPercent

ProtectionismData = ProtectionismData.dropna(subset =
    ['Unemployment']).reset_index(drop=True)

# Clean and add GDP per capita data

GDPCapitaData = pd.read_csv('GDPData.csv', skiprows = 4)

GDPCapita = []

for i in range(len(ProtectionismData)):
    code = ProtectionismData.loc[i, 'Country']
    year = str(ProtectionismData.loc[i, 'Year'])
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```
    row = GDPCapitaData[GDPCapitaData['Country Code'] == code]
    if len(row)>0 and year in GDPCapitaData.columns:
        GDPCapita.append(row[year].iloc[0])
    else:
        GDPCapita.append(np.nan)

ProtectionismData['GDP_PerCapita'] = GDPCapita

ProtectionismData = ProtectionismData.dropna(subset =
    ['GDP_PerCapita']).reset_index(drop=True)

# Clean and add GDP in US$ data

GDP_USData = pd.read_csv('GDP_US.csv', skiprows = 4)

GDP_US = []

for i in range(len(ProtectionismData)):
    code = ProtectionismData.loc[i, 'Country']
    year = str(ProtectionismData.loc[i, 'Year'])
    row = GDP_USData[GDP_USData['Country Code'] == code]
    if len(row)>0 and year in GDP_USData.columns:
        GDP_US.append(row[year].iloc[0])
    else:
        GDP_US.append(np.nan)

ProtectionismData['GDP_US'] = GDP_US

ProtectionismData = ProtectionismData.dropna(subset =
    ['GDP_US']).reset_index(drop=True)

# Clean and add inflation data

InflationData = pd.read_csv('InflationData.csv', skiprows = 4)

Inflation = []

for i in range(len(ProtectionismData)):
    code = ProtectionismData.loc[i, 'Country']
    year = str(ProtectionismData.loc[i, 'Year'])
    row = InflationData[InflationData['Country Code'] == code]
    if len(row)>0 and year in InflationData.columns:
        Inflation.append(row[year].iloc[0])
    else:
        Inflation.append(np.nan)

ProtectionismData['Inflation'] = Inflation

ProtectionismData = ProtectionismData.dropna(subset =
    ['Inflation']).reset_index(drop=True)

# Clean and add protectionism data
# NOTE: this is not world bank data, so structure is a bit different

PartyData = pd.read_csv('NoParties.csv', delimiter = ';', decimal = ',')

NoParties = []
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```
for i in range(len(ProtectionismData)):
    code = ProtectionismData.loc[i, 'Country']
    year = int(ProtectionismData.loc[i, 'Year'])

    match = PartyData[(PartyData['ccodealp'] == code) & (PartyData['year'] ==
        year)]

    if len(match) > 0:
        NoParties.append(match['gol_enep'].iloc[0])
    else:
        NoParties.append(np.nan)

ProtectionismData['NoParties'] = NoParties
ProtectionismData =
    ProtectionismData.dropna(subset=['NoParties']).reset_index(drop=True)

# Save everything to a single csv file

ProtectionismData.to_csv('ProtectionismDataset.csv')
```

B Preliminary DML Code

```
# Emily Harms
# POL380 Term Project: Causes of Protectionism
# Double Machine learning

# NOTES

# Oct 22
# Results differ every time (as usual for ML) --> Probably a good idea to do
    bootstrapping (I will add this for final paper, is ok for now since just
    proposal)

# Oct 28
# Remove number of political parties --> will focus on economic factors
# Need to figure out how to control for social interactions

# import required modules

import doubleml as dml
from sklearn.ensemble import RandomForestRegressor
import pandas as pd


# Import the data

data = pd.read_csv('ProtectionismDataset.csv')

data.pop('Country') # Ditch country and year since I don't want them impacting
    results
data.pop('Year')

# Define independent and dependent variables
```

```
dep_var = 'Tariff'

ind_vars = ['Exports', 'Unemployment', 'GDP_PerCapita', 'GDP_US', 'Inflation',
    'NoParties']

results = {} # dictionary for storage

# Loop through independent variables

for var in ind_vars:
    measured_var = var
    other_vars = [v for v in ind_vars if v != measured_var]

    dml_data = dml.DoubleMLData(data, y_col = dep_var, d_cols = measured_var,
        x_cols = other_vars) # Create data structure

    # define nuisance learners with Random Forest
    #Figure out what parameters make sense
    learner_l = RandomForestRegressor(n_estimators=100, max_depth=5,
        random_state=42) # predicting target from controls
    learner_m = RandomForestRegressor(n_estimators=100, max_depth=5,
        random_state=42) # predict treatment from controls

    # Partial Linear Regression Model
    dml_plr = dml.DoubleMLPLR(dml_data, ml_l = learner_l, ml_m = learner_m,
        n_folds = 5)
    dml_plr.fit()

    results[measured_var] = dml_plr.summary

# Combine results into a single DataFrame

resultsDF = pd.concat(results)

resultsDF.to_csv('DML_Results.csv')
```